Rabeya Akter

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Website · Google Scholar · GitHub · LinkedIn

Education

University of Dhaka Jan 2019 – Mar 2024
BSc in Robotics and Mechatronics Engineering Dhaka, Bangladesh
CGPA: 3.75/4.00 (Final year: 3.88/4.00)

Selected coursework: Introduction to Machine Learning; Digital Image Processing and Robot Vision; Digital Signal Processing; Object-Oriented Programming; Fundamentals of Programming; Linear Algebra; Multivariate and Vector Calculus; Statistics for Engineers.

Thesis: Continuous Bangla Sign Language Translation: Mitigating the Expense of Gloss Annotation with the Assistance of Graph. Thesis book

Research Focus

I study how intelligent agents can learn, reason, and act efficiently from multimodal information. My research spans reinforcement learning, world models, on-policy distillation, multimodal learning, and language model reasoning, with data efficiency and reliable generalization as unifying goals.

Publications & Preprints

[1] **Teach Yourself Where to Look: On-Policy Attention Self-Distillation for Reasoning**
S. H. Arib, **R. Akter**, I. N. Swapnil, Md F. A. Sayeedi, T. Mohiuddin, Md M. Islam
On-policy attention self-distillation improves mathematical reasoning accuracy by 4.98–8.40 percentage points.
Under review at ICLR 2027 · Paper

[2] **EK-BANGLA: When Hand–Object Interaction Meets Low-Resource Generation**
R. Akter, Md S. Shamil, A. K. M. F. Rabbi, A. Deb, R. Islam, Md A. Arefeen
Evaluates multimodal models on Bangla egocentric hand–object interactions through captioning and question answering.
Preprint · Project page

[3] **LAGEA: Language Guided Embodied Agents for Robotic Manipulation**
A. M. Chowdhury, A. M. R. Mazumder, S. H. Arib, **R. Akter**
Uses vision-language feedback on robot failures to shape reinforcement learning rewards for manipulation.
ICML 2026 · Paper

[4] **T3Time: Tri-Modal Time Series Forecasting via Adaptive Multi-Head Alignment and Residual Fusion**
A. M. Chowdhury, **R. Akter**, S. H. Arib
Aligns time, frequency, and language features for multivariate time series forecasting; 3.28% lower mean squared error.
AAAI 2026 · Paper

[5] **Beyond Visible Evidence: Open-World Amodal Counting Under Spatial Occlusion**
S. H. Arib, **R. Akter**, A. M. Chowdhury, Md J. A. Sourov, Md M. Hasan
Reconstructs occluded object features and aligns visual attention for robust open-world amodal counting.
WACV 2027 · Paper

[6] **SignFormer-GCN: Continuous Sign Language Translation Using Spatio-Temporal Graph Convolutional Networks**
S. H. Arib*, **R. Akter***, S. Rahman, S. Rahman (*equal contribution)
Combines video and skeletal features for gloss-free translation of Bangla, American, and German sign languages.
PLOS ONE · WiML Workshop, NeurIPS 2025 · Paper · Extended abstract

[7] **Bornil: An open-source sign language data crowdsourcing platform for AI enabled dialect-agnostic communication**
S. E. Dhruvo, M. A. Rahman, M. K. Mandal, Md I. H. Shihab, A. A. N. Ansary, K. F. Shithi, S. Khanom, **R. Akter**, S. H. Arib
Enables multilingual sign language data collection, annotation, and validation for low-resource recognition.
Preprint · Paper

[8] **Classical machine learning approach for human activity recognition using location data**
S. H. Arib, **R. Akter**, O. Shahid, Md A. R. Ahad
Recognizes eight locomotion and transport activities from smartphone location features using random forests.
Adjunct Proceedings of UbiComp / ISWC 2021 · Paper · Poster

Research Experience

Research Collaboration Mar 2026 – Present
Collaborators: Dr. Md Mofijul Islam (Amazon GenAI, USA); Dr. Tasnim Mohiuddin (QCRI, Qatar) Remote

- Co-developed solution-conditioned attention supervision for **on-policy self-distillation**; improved Qwen3 mathematical reasoning by **4.98–8.40 percentage points** over token-only self-distillation with **72.6% less estimated model computation**.

Research Collaboration Sep 2025 – Present
Collaborator: Dr. Ser-Nam Lim (University of Central Florida, USA) Remote

- Working on **world models** for reinforcement learning and **on-policy distillation** for language model reasoning.

North South University Mar 2026 – Present
Research Assistant Dhaka, Bangladesh
Department of Electrical and Computer Engineering

Supervisors: Md Salman Shamil and Dr. Md Adnan Arefeen

- Developed **EK-Bangla** for egocentric hand–object understanding, with **9,600 human-verified Bangla captions** and **2,000 multiple-choice questions**; evaluated **nine multimodal language models** to characterize language-dependent reasoning gaps.

University of Dhaka Jun 2025 – Feb 2026
Research Assistant Dhaka, Bangladesh
Department of Robotics and Mechatronics Engineering

Supervisor: Dr. Md Mehedi Hasan

- Developed **CountOCC**, reconstructing occluded object features with multimodal guidance and attention consistency; reduced mean absolute counting error by **20.80%** on FSC-147-OCC test and **49.89%** on CARPK-OCC over prior baselines.

University of Dhaka Jan 2023 – Dec 2023
Undergraduate Research Assistant Dhaka, Bangladesh
Department of Robotics and Mechatronics Engineering

Supervisors: Dr. Shafin Rahman and Dr. Sejuti Rahman

- Developed **SignFormer-GCN**, fusing video and skeletal representations for gloss-free sign language translation; achieved **19.75 BLEU-4** on RWTH-PHOENIX-2014T and **8.53 BLEU-4** on How2Sign test with **9.43 million parameters**.

Professional Experience

Therap (BD) Ltd. Apr 2024 – May 2025
Associate Software Engineer, QA – Machine Learning Team Dhaka, Bangladesh

- Researched Amazon Alexa devices and sound detection systems; reviewed action recognition datasets and contributed to data collection design.
- Led quality assurance for a video redaction system, developing comprehensive test plans and cases to validate machine learning algorithms used in video processing.

Pathao Limited Jan 2024 – Mar 2024
AIM Intern – Data Science, Analytics and Insights Team Dhaka, Bangladesh

- Developed five key performance indicators and metrics to measure Pathao Courier brand health, aligned with company-wide strategic goals.
- Designed and implemented questionnaires using data-driven methodologies to assess brand health, improving response quality and depth of insights.

Teaching Experience

Bangladesh Artificial Intelligence Olympiad Jun 2024 – Aug 2024
Instructor, National Camp Bangladesh

- Designed and delivered lessons on classification algorithms and evaluation metrics for high school students; created quizzes to assess their understanding.

Selected Academic Projects

Automated Stock Trading Code — Aug 2022 – Nov 2022
Supervisor: Dr. Sejuti Rahman · *Reinforcement Learning*

Human Activity Recognition Using Location Data Code — Apr 2021 – Jun 2021
Supervisor: Dr. Md Atiqur Rahman Ahad · *Machine Learning*

Comparative Analysis of Emotional Support Delivery by NAO Robot and Humans Across Varied Emotional States Code · Demo — Aug 2023 – Nov 2023
Supervisor: Dr. Lafifa Jamal · *Human–Robot Interaction*

Technical Skills

Methods	Reinforcement learning, on-policy self-distillation, multimodal learning, language model post-training, computer vision, time series forecasting
Programming	Python
ML frameworks	PyTorch, Scikit-learn
Agents & tools	LangChain, FastAPI, Docker, Git
Data processing	Pandas, NumPy, OpenCV, SciPy

Honors & Academic Activities

- **Reviewer**, AAAI 2026
- **Special Innovation Fund Research Grant**, Ministry of Science and Technology (SRG-232431) — 2023–2024
- **Engineering Faculty Undergraduate Merit Scholarship**, University of Dhaka — 2024
- **Dhaka Board Talent Pool Scholarship**, Higher Secondary Certificate Examination — 2018
- **Honourable Mention**, NASA Earth Observatory Dashboard Hackathon — 2021
- **11th place**, Sussex Huawei Locomotion Challenge — 2021
- **Second Runner-up**, Ada Lovelace Datathon — 2021
- **10th place**, 3rd Women's Mathematics Olympiad — 2019
- **2nd place**, Pi Day Celebration — 2019
- **Student Activity Secretary**, IEEE, University of Dhaka — 2023
- **Mentor**, Bangladesh Robot Olympiad (BdRO); school robotics workshops — 2019–2020
- **Speaker, Dui Prishthar Bisheshoggo**, IEEE Women In Engineering — 2023

References

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Dr. Md Mehedi Hasan
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